

Summary so far

Classical phylodynamics:

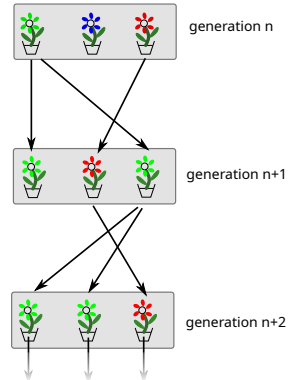
- Coalescence for pathogens alone
- time continuous (ODE model)
- “birth rate”: Incidence
- “population size”: prevalence
- time scale separation from host coalescence

Our approach:

- Host and pathogen coalescent intertwined
- Host: separated generations (Wright-Fisher model)
- Pathogen: Vertical transmission from one generation to the next
- Pathogen: SI-model within one host generation.

Recap: Wright-Fisher Model (Host)

- Fixed population size N
- Non-overlapping generations
- each “daughter” randomly selects a “mother”
- eventually we have ascendants of only one individual (fixation)

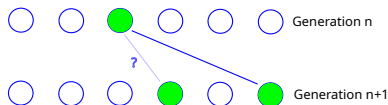


Host coalescent, sample size 2

Sample of 2 individuals:

- population size N
- we go one step **backward in time**
- Coalesc. prob. (prob. for identical mother)

$$p_c = 1/N$$



Time to host coalescent, sample size 2

Sample of 2 individuals:

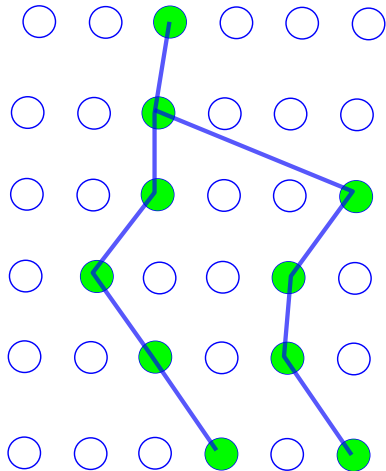
Coalesc. prob. $p_c = 1/N$

Coalesc after exactly generations:

$K \sim \text{Geom}(p_c)$

Expected time to coalesc.:

$E(K) = 1/p_c = N$



Estimation of population size N

Sample of 2 individuals:

Mutation-rate u per generation,

Number of mutations S ,

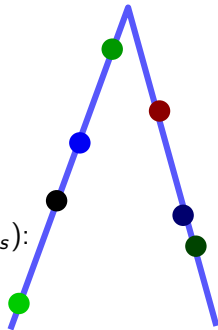
$$S \sim \text{Pois}(2Ku) \quad E(S) = 2uE(K) = 2uN$$

Moment estimator for N (No observed mutations S_{obs}):

$$\hat{N} = S_{obs}/(2u)$$

Sample size > 2 :

- (a) Multiple binary estimations
- (b) Watterson-Estimator.



Dynamics of Pathogens

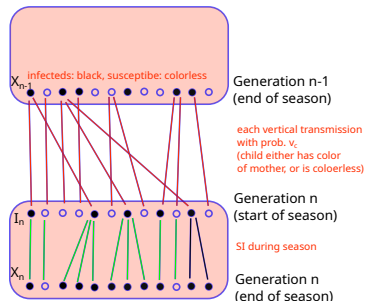
End of last season: Prevalence/number of infecteds
 X_{n-1} (given)

Beginning of season, founding infecteds:
 $I_n \sim \text{Bin}(X_{n-1}, c_v)$

Within-season, SI process:

$J_0 = I_n$, $J_t \rightarrow J_t + 1$ rate $\beta J_t (N - J_t)$

End of season, X_n of this season
 $X_{n+1} = J_G$, G : duration of a season



Deterministic Dynamics of Pathogens

X_{n-1} (no of infecteds, end of season) given

$$x_{n-1} = X_{n-1}/N$$

$I_n \sim \text{Bin}(X_{n-1}, c_\nu)$ founding infecteds

$$i_n = I/N \rightarrow c_\nu x_n \text{ for } N \rightarrow \infty$$

$J_t \rightarrow J_t + 1$ rate $\beta J_t(N - J_t)$ SI process

$$j(t) = J_t/N, \quad j' = \beta j(1-j), \quad j(0) = i_n = c_\nu x_n$$

$$j(t) = \frac{i_n}{e^{-\beta t}(1 - i_n) + i_n}$$

$$x_{n+1} = j(G) = \frac{i_n}{e^{-\beta G}(1 - i_n) + i_n} = \frac{c_\nu x_n}{e^{-\beta G}(1 - c_\nu x_n) + c_\nu x_n}$$

Stationary state of Pathogens

$$x_n = x_{n+1} = x$$

$$x = \frac{c_v x}{e^{-\beta G}(1 - c_v x) + c_v x}$$

- Solve for x (endemic equilibrium):

$$x = x^* = \frac{c_v - (1 - c_v) e^{-\beta G}}{c_v^2 + (1 - c_v) c_v (1 - e^{-\beta G})}$$

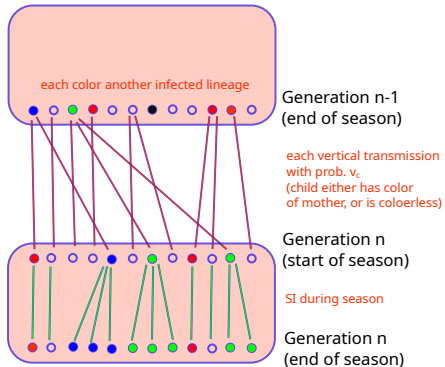
- Solve for $e^{-\beta G}$ (for estimation, given c_v and x^*):

$$e^{-\beta G} = \frac{c_v}{(1 - c_v)} \frac{(1 - x^*)}{(1 - c_v x^*)}.$$

(for $x^* \in (0, 1)$ and $0 < c_v < c_v^{crit} < 1$)

Coalescence of two pathogen lineages

Select randomly two different infected individuals, and check if they have the same parent/same color



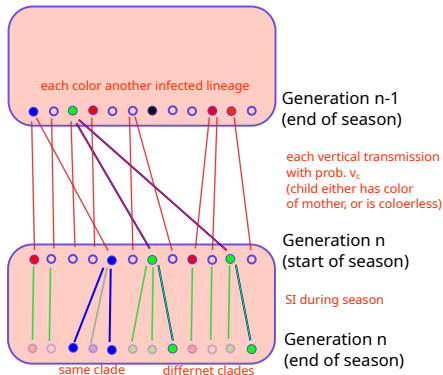
2 possibilities for coalescence event

Same clade/founding infecteds

- same clade
- name this probability π

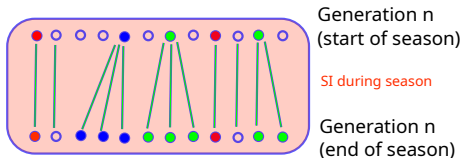
Different clades/founding infecteds

- different clades
- probability $(1 - \pi) \frac{1}{N}$



Coalescence for the SI process: Polya Urn

- We start with I founder infecteds/different colors
- Newly infecteds are produced
- We draw two infecteds; $\pi = P(\text{same color})$



If we add one newly infecteds:

$$P(\text{color}) = \frac{\text{number of infecteds of this color}}{\text{number of infecteds}}$$

Coalescence for the SI process: Setup

Polya-Urn (epidemic urn):

- start with k balls, each with another color
- draw a ball
- put this ball back, plus another ball of the same color
- do that for n steps.

Self-Reinforcing

Theorem (well known)

Let $1 + x_i$ the number of balls with the i 'th color after n steps, such that $(x_1, \dots, x_n)$ is a partition of n .

Then, the probability is identical for all partitions $(x_1, \dots, x_n)$ of n .

Coalescence for the SI process

We take a Polya Urn with k colors after n steps.

$$\pi = P(\text{two balls drawn without replacement have the same color})$$

Theorem

$$\pi = \frac{n}{n+k-1} \frac{2}{k+1}$$

Proof: Let $C = (1 + x_1, \dots, 1 + x_k)$ with $\sum_{\ell=1}^k x_{\ell} = n$ in the urn, then

$$\pi|C = \sum_{\ell=1}^k \frac{\ell}{(k+n)} \frac{\ell-1}{(k+n-1)}.$$

Remove condition (all partitions have identical probability)

$$\pi = \sum_C (\pi|C) P(C) = \frac{n}{n+k-1} \frac{2}{k+1}$$



C Side remark

$$\pi = \frac{n}{n+k-1} \frac{2}{k+1}$$

Case: Few colors (k finite), many new balls (n large)

$$\pi = \frac{n}{n+k-1} \frac{2}{k+1} \rightarrow \frac{2}{k+1} \quad \text{for } n \rightarrow \infty$$

Naive idea: Many balls, drawing with and without replacement identical.

Symmetry: Each color has probability $1/k$.

$P(\text{both balls have the same color}) = \frac{1}{k}$

BUT: $\pi = \frac{2}{k+1} > \frac{1}{k}$

$\Rightarrow$ Self-Reinforcement at work.

Coalescence for the SI process

$$\pi = \frac{n}{n+k-1} \frac{2}{k+1}$$

k : Founder infected cases:

$$k = I = c_v x^* N$$

n : Newly infecteds = Infecteds - founder infecteds

$$n = x^* N - c_v x^* N = (1 - c_v) x^* N.$$

$$\pi = \frac{n}{n+k-1} \frac{2}{k+1} = \frac{(1 - c_v) x^* N}{(1 - c_v) x^* N + c_v x^* N - 1} \frac{2}{c_v x^* N + 1} \approx \frac{2(1 - c_v)}{c_v x^*} \frac{1}{N}$$

Coalescence for the pathogens

$$p_i = \frac{2(1 - c_v)}{c_v X^*} \frac{1}{N}$$

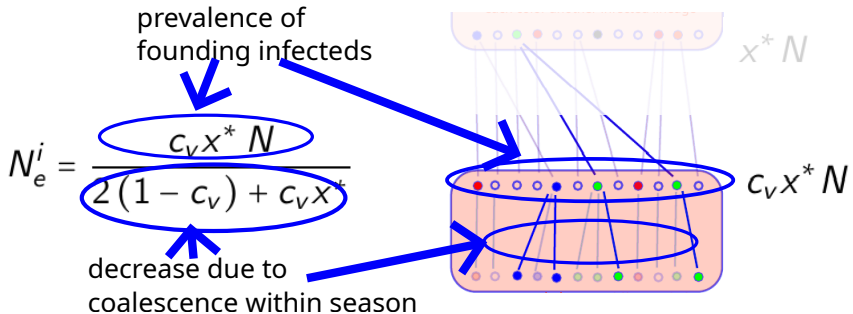
Coalescent probability for two pathogen lineages in one generation:

$$p_c^i = \pi + (1 - \pi) \frac{1}{N} \approx \pi + \frac{1}{N} = \frac{2(1 - c_v) + c_v X^*}{c_v X^*} \frac{1}{N} = \left(1 + \frac{2(1 - c_v)}{c_v X^*}\right) \frac{1}{N}$$

Find **effective population size** N_e^i such that $p_c^i = 1/N_e^i$:

$$N_e^i = \frac{c_v X^*}{2(1 - c_v) + c_v X^*} N.$$

Coalescence for the pathogens



Parameter estimation

- get a sample of hosts
- determine x^* as the fraction of infecteds
- determine N from the hosts's coalescence (Watterson estimator)
- determine N_e^i from the pathogen coalescence (Watterson estimator)
- determine c_v from x^* , N und N_e^i via

$$N_e^i = \frac{c_v x^*}{2(1 - c_v) + c_v x^*} N.$$

- determine βG via

$$e^{-\beta G} = \frac{c_v}{(1 - c_v)} \frac{(1 - x^*)}{(1 - c_v x^*)}.$$

Conclusions

Achievements:

- Wright-Fisher model with embedded time-continuous SI model
- Coalescence of pathogens determined
- Rather a research platform than a final result

Questions

- Coalescent tree (Pathogen) embedded in coalescent tree (Host)
- Reduction of variance as $\text{cov}(N, N_e^i) > 0$?

Replace SI by SIS or SIR. Multiple merger?

- Multi-type (resistance, virulence) possible to analyze?